FedAvg vs. FedProx on PathMNIST: 5-Seed Empirical Benchmark

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1. Research Objective & Setup

Evaluates whether **FedProx** (μ = 0.01) mitigates local parameter drift and improves test generalization compared to **FedAvg** on histopathological imaging under simultaneous non-IID partitioning and 40% symmetric label noise.

- **Dataset:** PathMNIST (89,996 train, 10,004 val, 7,180 test; 9 tissue classes).
- **Heterogeneity:** 5 clients, Dirichlet class skew (α = 0.3).
- **Label Noise:** Client 5 trained on exactly **40.00% symmetric noise**.
- **Model:** CompactPathCNN (2 Conv2D, AdaptiveAvgPool, 2 Linear; **36,425 parameters**).

2. Execution & Protocol Controls

Seed-controlled paired benchmark across 5 seeds (**42, 123, 2026, 3407, 7777**). 30 rounds per algorithm (300 communication rounds total) on 2 PyTorch CPU threads (AMD Ryzen 5 5600H).

- **Hyperparameters:** Local epochs E=2, batch size B=32, lr η=0.01.
- **Purity Controls:** Seed-matched initial weights, paired partitions.
- **Quarantined Evaluation:** Test set evaluated strictly post-training.
- **Compute Time:** 471.7 min total (Avg 42.9m/seed, Prox 51.5m/seed).

3. Quantitative Results Across All 5 Seeds (Held-Out Test Set)

Seed	Client 5 Noise	FedAvg Test Acc	FedProx Test Acc	Paired Δ (pp)	FedAvg Macro-F1	FedProx Macro-F1	Δ Macro-F1	FedAvg Bal Acc	FedProx Bal Acc
42	3,158 / 7,895 (40%)	79.62%	79.64%	+0.01 pp	0.7403	0.7413	+0.0010	74.36%	75.43%
123	8,381 / 20,952 (40%)	70.74%	70.29%	-0.45 pp	0.6181	0.6136	-0.0045	64.79%	64.32%
2026	8,054 / 20,134 (40%)	57.73%	57.98%	+0.25 pp	0.5186	0.5133	-0.0053	56.29%	56.31%
3407	5,292 / 13,231 (40%)	75.20%	71.67%	-3.52 pp	0.6214	0.5957	-0.0257	65.63%	63.18%
7777	8,272 / 20,680 (40%)	61.11%	61.81%	+0.70 pp	0.5057	0.5030	-0.0027	54.80%	55.28%
Mean ± Std	—	**68.88% ± 9.27%**	**68.28% ± 8.55%**	**-0.60 ± 1.69 pp**	**0.6008 ± 0.095**	**0.5934 ± 0.096**	**-0.0075 ± 0.010**	**63.18% ± 7.92%**	**62.90% ± 8.07%**

Statistical Inference (df = 4): Test Accuracy paired t(4) = -0.7983, **p = 0.46938** (95% CI: [-2.69 pp, +1.49 pp]; Wilcoxon p = 1.000). Macro-F1 paired t(4) = -1.5908, **p = 0.18686** (95% CI: [-0.0205, +0.0056]). *Interpretation:* **No statistically significant difference in test performance was detected between FedAvg and FedProx across the five evaluated seeds.** Because five seeds provide limited statistical power and no equivalence testing was conducted, this does not establish statistical equivalence.

4. Core Insights & Research Recommendations

- **Symmetric Noise vs. Proximal Penalty:** FedProx penalizes parameter deviation from the global model ($\mu/2 \, ||w - w^t||^2$) but does *not* correct corrupted gradient orientations induced by mislabeled samples. Client 5's corrupted update is still integrated proportionally into the global model during sample-weighted aggregation.
- **Peak Validation Accuracy:** Peak validation accuracy favored FedAvg by -0.52 pp (p = 0.0022 across 5 seeds); we report this empirical difference directly without asserting an unsupported causal claim.
- **Bounded Finding:** Within the evaluated configuration (PathMNIST, 5 clients, α=0.3, Client 5 40% noise, CompactPathCNN, FedProx μ=0.01), proximal regularization did not yield generalization benefits over FedAvg. Defending against label noise motivates investigating explicit noise-aware paradigms (e.g., robust losses, cosine filtering, or local sample pruning).